PAPER PRESENTATION

Programming Refusal with Conditional Activation Steering

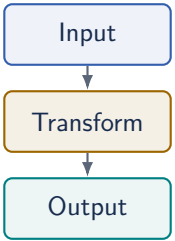
Conditional control of model behavior

Tia Manoukian, Nada Hammad, Nikola Dobričić

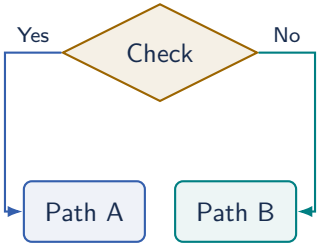
Paper by Bruce W. Lee et al. | ICLR 2025

Example slide: visual style guide

Process



Decision



Geometry

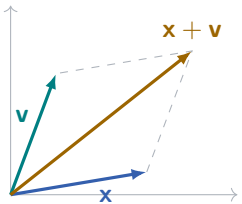


Chart (synthetic values)

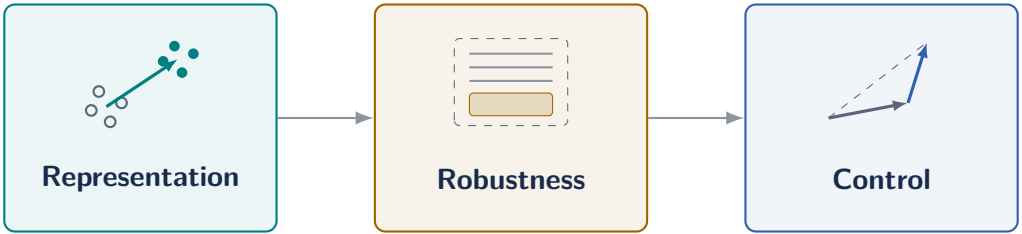


Example formulas

$$\mathbf{z} = \mathbf{x} + \alpha\,\mathbf{v}$$

$$\cos\theta = \frac{\mathbf{x}^{\mathsf{T}}\mathbf{v}}{\|\mathbf{x}\|\,\|\mathbf{v}\|}$$

Does finding a direction mean we can control behavior?



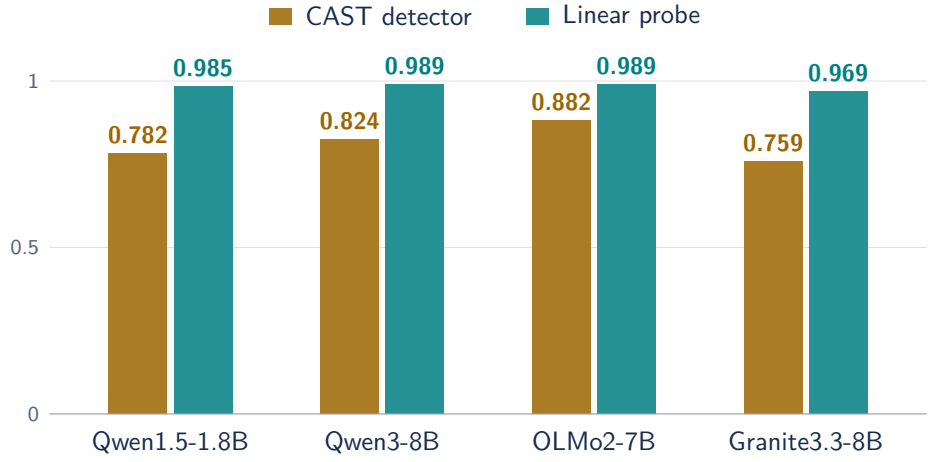
Can we detect the concept?

Does detection survive context changes?

Can we change model behavior?

Our experiments, not results from the paper | Qwen1.5-1.8B, Qwen3-8B, OLMo2-7B, Granite3.3-8B

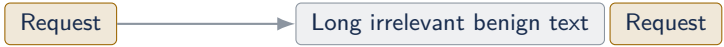
The information is there — but CAST doesn't extract all of it



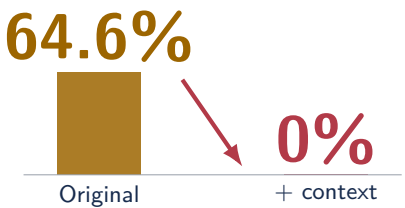
Held-out F1 · Same hidden-state features

Our experiments, not results from the paper | seed means over three family splits

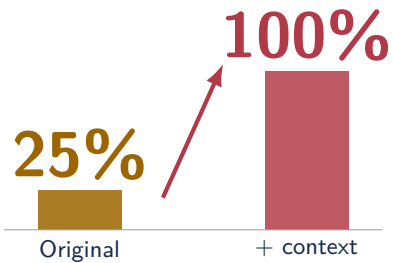
A harmless change in context can break the detector



Qwen1.5-1.8B Harmful prompts



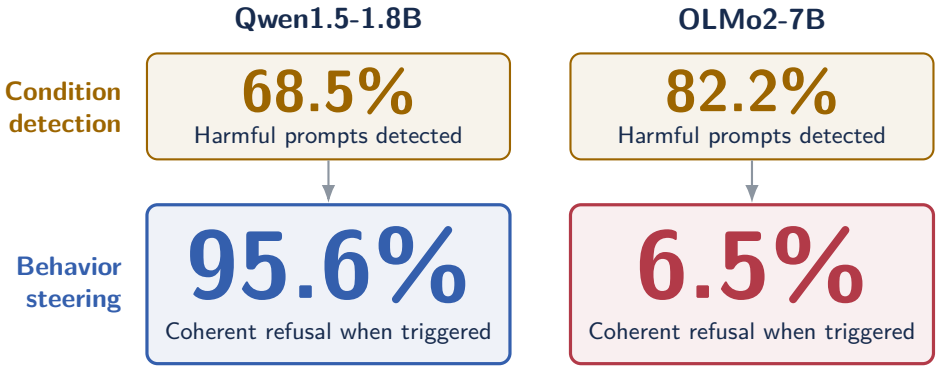
Qwen3-8B Benign prompts



Opposite failures. Same loss of selectivity.

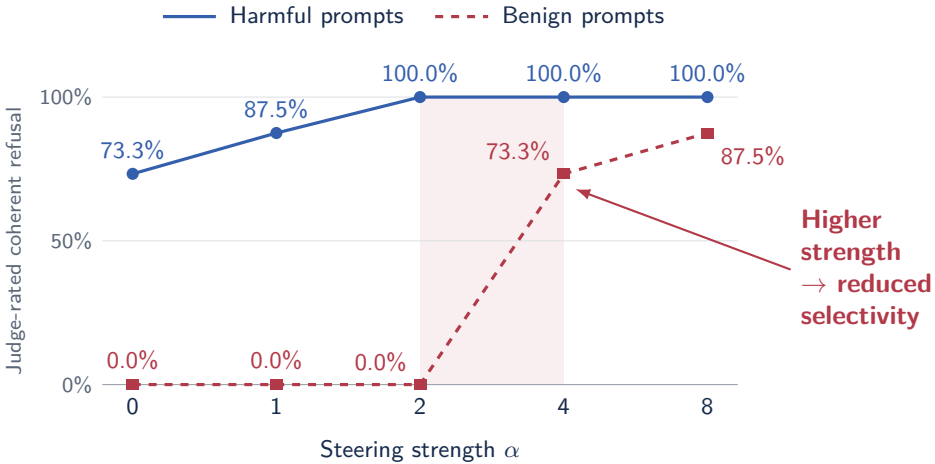
Our experiments, not results from the paper | gate activation, not refusal · 16 + 16 families × 3 seeds

Detecting a condition is not the same as steering behavior



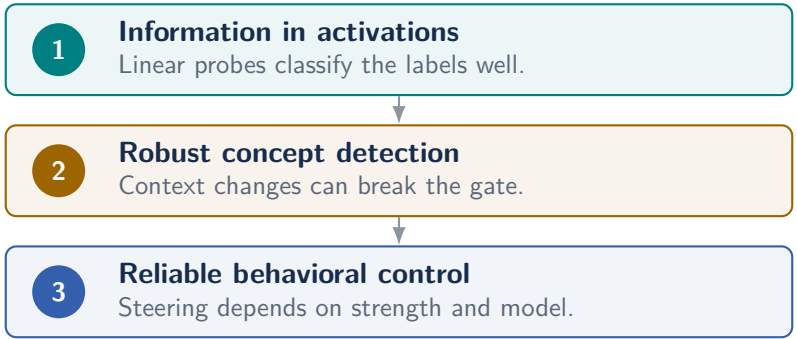
Our experiments, not results from the paper | main test · coherent refusal: model-judge rating

More steering is not necessarily better



Our experiments, not results from the paper | Qwen3-8B, no gate · 16 + 16 families × 3 seeds

Three different problems — not one



A decodable representation is not necessarily a robust detector or an effective intervention.

Our experiments, not results from the paper | Qwen1.5-1.8B, Qwen3-8B, OLMo2-7B, Granite3.3-8B

Does CAST need its nonlinear similarity function?

BACKUP

Qwen1.5-1.8B · held-out F1 · Float64 evaluation · Fixed condition layer

CAST similarity

$$\cos\big(\mathbf{h},\ \tanh(\mathrm{proj}_{\mathbf{c}}\,\mathbf{h})\big)$$

0.619

F1

Direct cosine

$$\cos(\mathbf{h}, \mathbf{c})$$

0.736

F1

CAST similarity: Lee et al., ICLR 2025, Sec. 3.1. | F1: our experiments, seed means.